

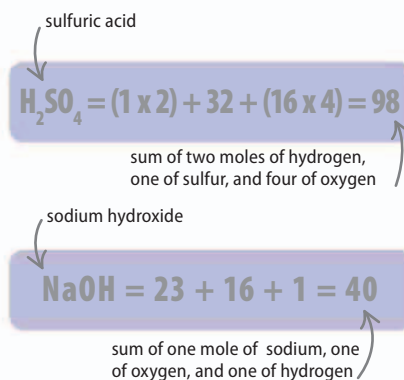
Moles

Chemists measure quantities of reactants and products in moles. A mole is a standard unit of atoms. It is defined as the number of atoms in 12 g (0.5 oz) of carbon. This mass is known as the relative atomic mass (RAM) of carbon. A mole of anything else contains this same number of atoms (roughly 6.02×10^{23}), but because atoms have different masses, a mole of one element will have a different mass from a mole of another. Compounds have relative formula masses (RFM), which are calculated by adding up the RAM of their constituents.

Element	RAM
hydrogen	1
carbon	12
oxygen	16
sodium	23
sulfur	32
iron	56
gold	197

△ Relative atomic mass

Atoms of different elements have different masses, so moles of different substances have different masses too. For example, one mole of carbon weighs twelve times more than one mole of hydrogen.



△ Relative formula mass

The RFM of a compound is the sum of the RAM of each of its constituent atoms.

Apparatus diagram

Chemists often draw the apparatus they use in experiments so that other scientists can replicate the equipment. The apparatus is drawn in a simple, two-dimensional style that makes each item easy to recognize. Test tubes, beakers, conical flasks, and other items of glassware are used to hold reactants during heating. They are frequently positioned on a steel tripod that holds them above the Bunsen burner.

